

8) Solve the eqⁿ $\frac{\partial u}{\partial t} = 1 \frac{\partial^2 u}{\partial x^2}$ with

boundary cond^{ns}.

$$u(x, 0) = 3 \sin n \pi x, \quad 0 \leq x \leq 1.$$

$$u(0, t) = 0 \quad \forall t$$

$$u(1, t) = 0 \quad \forall t$$

where $u(x, t)$ is finite.

Find $u(x, t)$ at any time t .

9) Solve the eqⁿ. $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$,

subject to following cond^{ns}.

$$i) u(x, 0) = u_0 \sin^3 \left(\frac{\pi x}{l} \right), \quad \forall x \in (0, l).$$

$$ii) u(0, t) = 0 \quad \forall t$$

$$iii) u(l, t) = 0 \quad \forall t$$

$$iv) u(x, \infty) \text{ is bounded.}$$

Find $u(x, t)$ at any time t .

10) A homogeneous rod of conducting material of length " l " has its ends kept at zero temperature and temperature of the rod initially is $u(x, 0) = a \sin \frac{\pi x}{l}$.

Find the temperature $u(x, t)$ at any time t .

4) i) $u(x, t)$ is bounded

ii) $u(0, t) = 0 \quad \forall t$

iii) $u(l, t) = 0 \quad \forall t$

iv) $u(x, 0) = \frac{u_0 x}{l} ; 0 \leq x \leq l$.

5) i) $u(x, t)$ is bounded

ii) $u(0, t) = 0 \quad \forall t$.

iii) $u(1, t) = 0 \quad \forall t$

iv) $u(x, 0) = x(1-x) ; 0 \leq x \leq 1$.

6) A homogenous rod of conducting material of length 4m has its ends kept at zero temperature. and the temperature initially is,
 $u(x, 0) = 2x ; 0 \leq x \leq 4$.
Find the temperature $u(x, t)$ at any time

7) Solve the equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$,

Subject to the cond^{ns}:

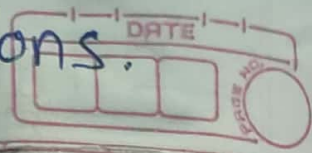
i) $u(x, 0) = 1 ; 0 < x < \pi$.

ii) $u(0, t) = u(\pi, t) = 0 \quad \forall t$.

iii) $u(x, t)$ is finite.

find $u(x, t)$. at any time t

AIDS.
Practice Questions.
Unit - 2.



* Examples on One dimensional Heat eqⁿ:-

* Solve the one-dimensional heat flow equation.

$$\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} \text{ for the function}$$

$u(x, t)$ subject to following cond^{ns}.

- 1) i) $u(0, t) = 0 \quad \forall t$
ii) $u(l, t) = 0 \quad \forall t$
iii) $u(x, 0) = x \quad ; \quad \forall x \in (0, l)$
iv) $u(x, \infty)$ is finite.

- 2) i) $u(0, t) = 0 \quad \forall t$
ii) $u(20, 0) = 0 \quad \forall t$
iii) $u(x, 0) = 20x - x^2, \quad \forall x \in (0, 20)$
iv) $u(x, t)$ is finite.

- 3) i) $u(0, t) = 0 \quad \forall t$
ii) $u(\pi, t) = 0 \quad \forall t$
iii) $u(x, 0) = \pi x - x^2, \quad 0 \leq x \leq \pi$
iv) u is finite $\forall t$.